

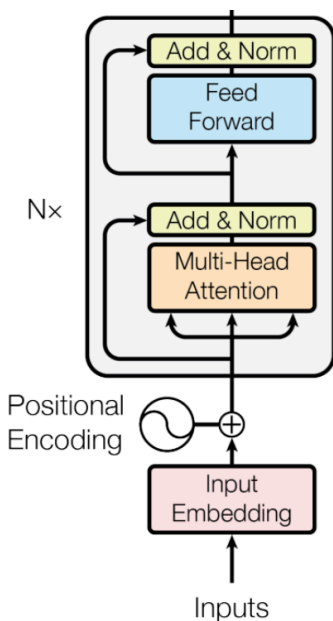
TRANSFORMERS

AI506 — Advanced Machine Learning

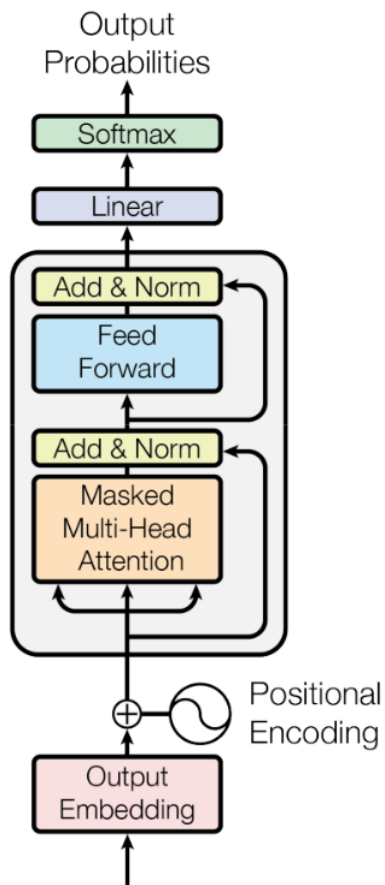
Simon Holm

May 2026

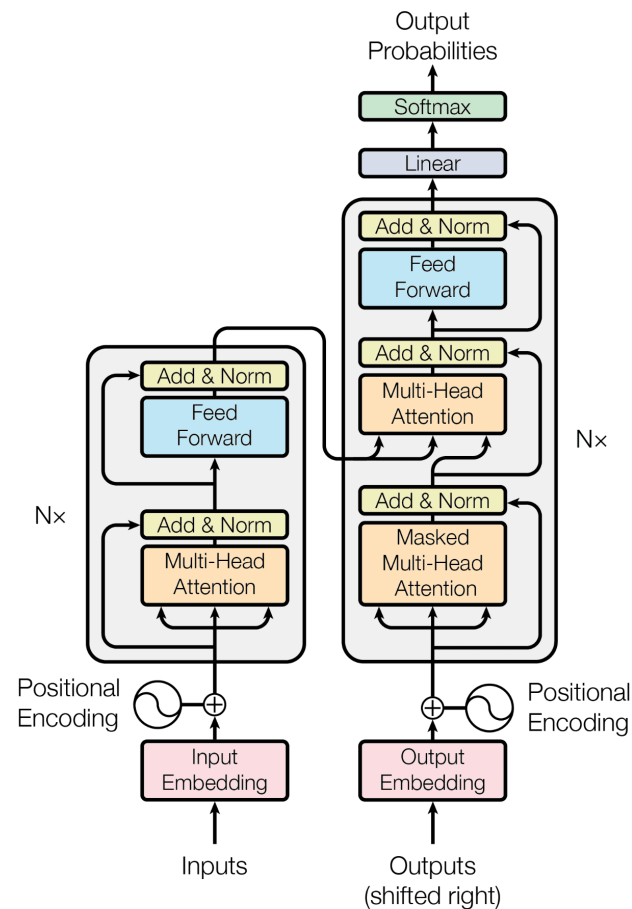
Encoder

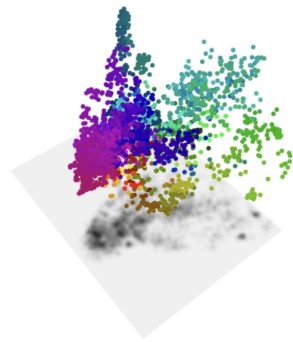


Decoder

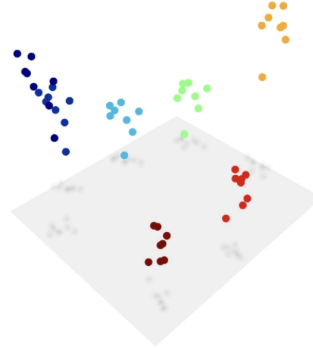


Encoder-Decoder





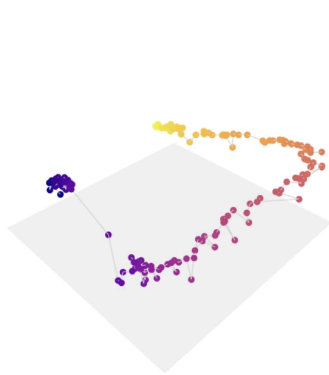
GEOGRAPHY



DAYS OF THE WEEK



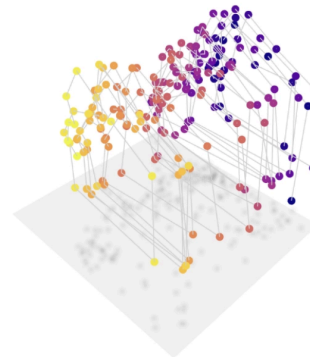
FORMALITY



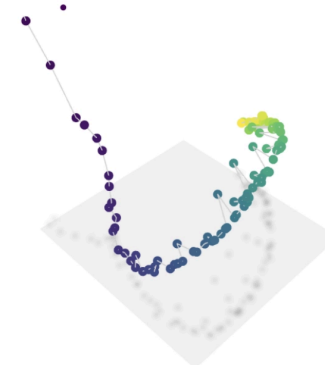
TEMPERATURE



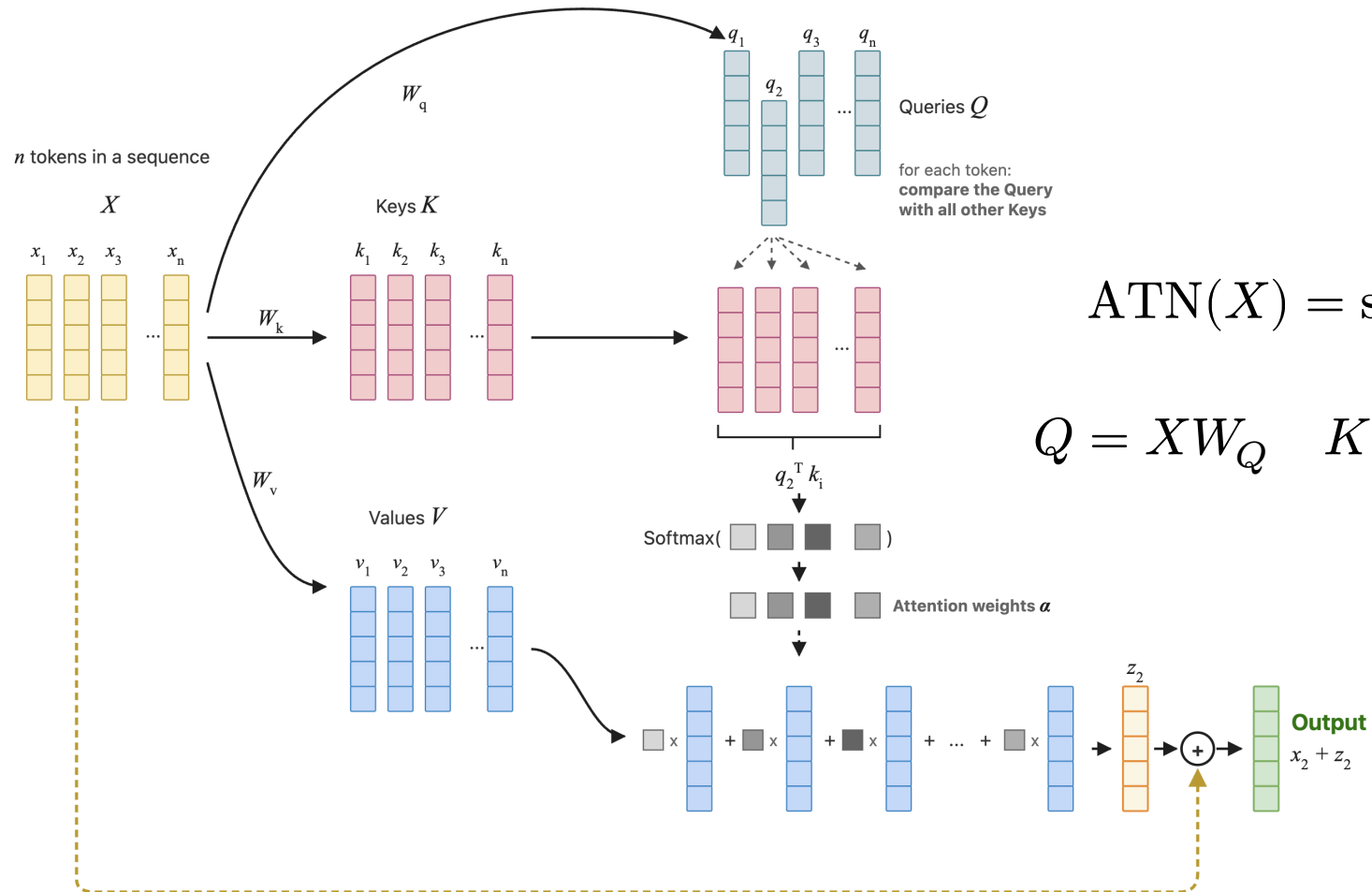
COLOR



YEARS

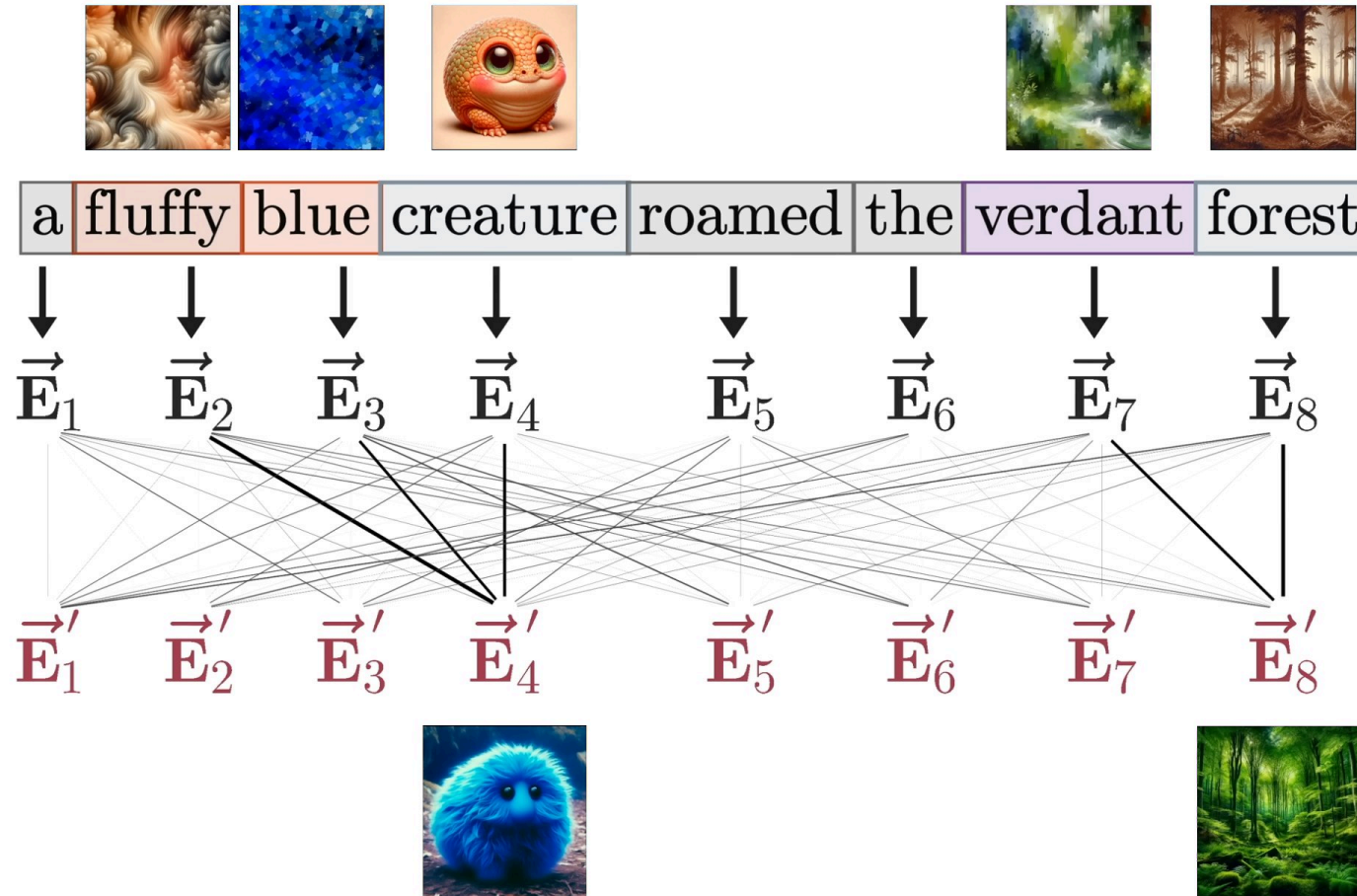


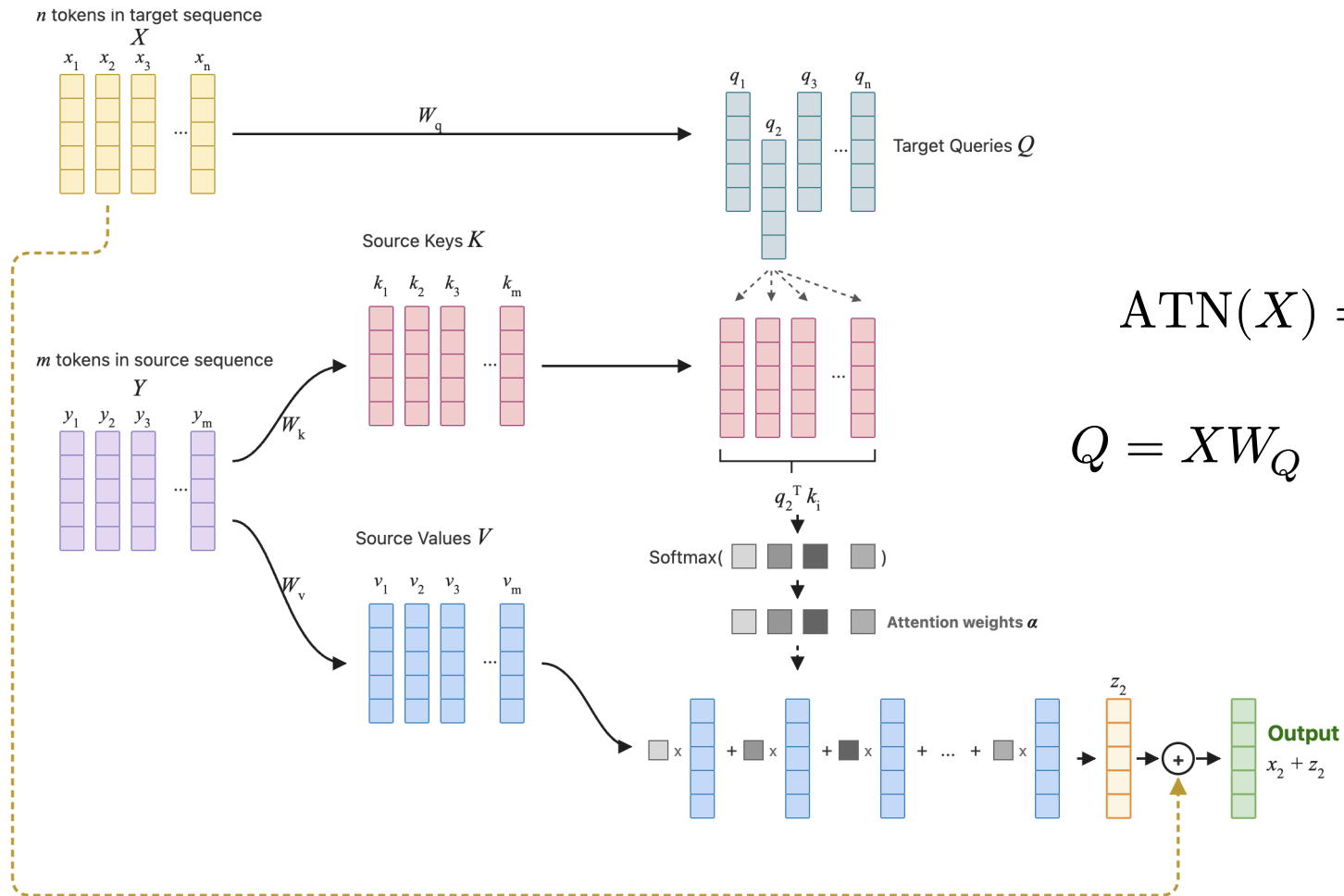
AGE



$$\text{ATN}(X) = \text{softmax}\left(\frac{Q^T K}{\sqrt{d_k}}\right) V$$

$$Q = XW_Q \quad K = XW_K \quad V = XW_V$$



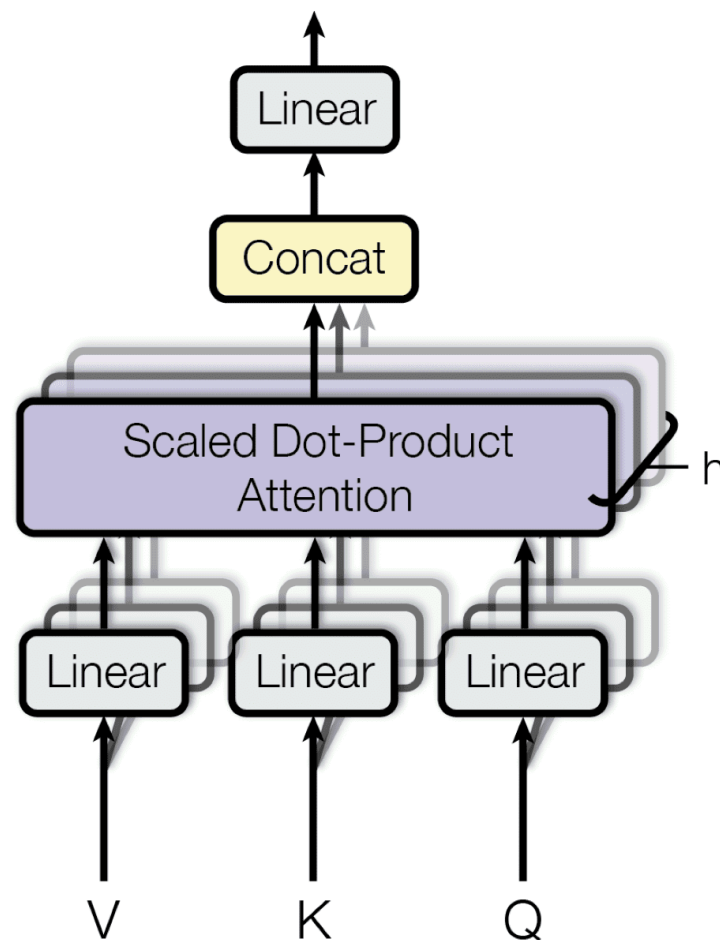


$$\text{ATN}(X) = \text{softmax}\left(\frac{Q^T K}{\sqrt{d_k}}\right) V$$

$$Q = XW_Q \quad K = ZW_K \quad V = ZW_V$$

$$\text{MultiHead}(X) = \text{head}_1 \uplus \dots \uplus \text{head}_h$$

$$\text{head}_i = \text{softmax}\left(\frac{Q^T K}{\sqrt{d_k}}\right) V$$



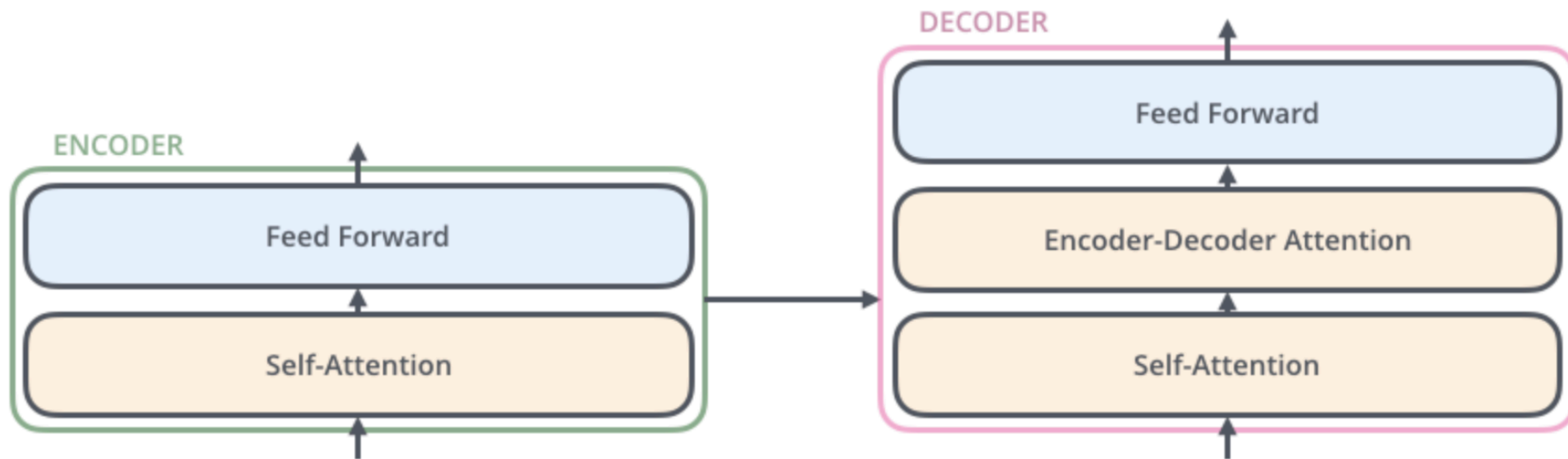
- Absolute position

$$v_i = W_v \cdot (x_i + p_i)$$

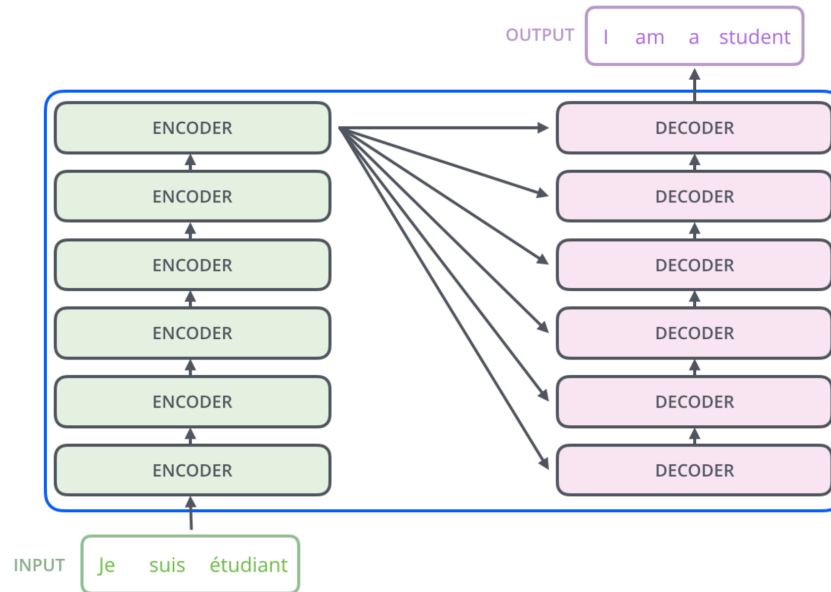
- Relative position

$$q_m^T k_n = x_m^T W_Q^T W_K x_n + b_{n-m}$$

		"Dream" x(1)	"big" x(2)	"and" x(3)	"work" x(4)	"for" x(5)	"it" x(6)
"Dream"	x(1)	α_{11}	α_{12}	α_{13}	α_{14}	α_{15}	α_{16}
"big"	x(2)	α_{21}	α_{22}	α_{23}	α_{24}	α_{25}	α_{26}
"and"	x(3)	α_{31}	α_{32}	α_{33}	α_{34}	α_{35}	α_{36}
"work"	x(4)	α_{41}	α_{42}	α_{43}	α_{44}	α_{45}	α_{46}
"for"	x(5)	α_{51}	α_{52}	α_{53}	α_{54}	α_{55}	α_{56}
"it"	x(6)	α_{61}	α_{62}	α_{63}	α_{64}	α_{65}	α_{66}



$$\text{FFN}(x) = W_2 \cdot \text{GeLU}(W_1 x + b_1) + b_2$$



$$h^{(l+1)} = \text{LN}(h^{(l)} + \text{FFN}(h^{(l)}))$$

$$\text{LN}(x) = \frac{x - \mu}{\sqrt{\sigma^2 + \varepsilon}}$$